INSTALLATION GUIDE

DIRECTED®



403.FRD25EU v1.00

2018 Ford C-Max (Smart Key) EU

Functions compatible with the 2018 Ford C-Max (Smart Key) EU

- Brake Status (foot brake)
- Clutch Activation
- Control of aftermarket alarm with OEM remote
- Door Lock Control
- Door Unlock
- E-Brake Status
- Entry Monitoring ALL Door Pins
- Entry Monitoring Trunk/Hatch Pin
- EU compatible (DB3/DS4)
- Ignition Status
- Parking lights via hazard control
- RAP Shut Down (Retained ACC Power)
- Tach / RPM Output
- Trunk / Hatch Release
- Universal RF Bypass- Key Req'd

This product is intended for installation by a professional installer only! Attempts to install this product by a person other than a trained professional may result in severe damage to a vehicle's electrical system and components.

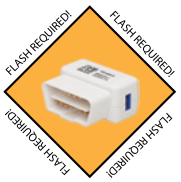
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Introduction

DB3 is an all-in-one door lock and override module.



Warning!

This module can only be flashed and configured using DirectLink at www.directechs.com or using the Directechs Mobile application for smartphones. Refer to "Connecting the module" for more information.

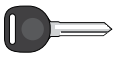
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Pre-installation and application warnings



This section highlights important information for this specific firmware and will assist in pricing accordingly, as well as bringing awareness to any operational or vehicle limitations.

Compatibilities and Requirements

T-Harness compatible 	T-Harness is not compatible.		**Keys required for programming**	**1**	**Keys required for operation** **1**

Vehicle Takeover **No** ☒ The engine will shut down as soon as a door is opened.

Firmware notes

- **Transponder override:** This firmware does not support transponder override. One (1) programmed OEM key must be wired into the vehicle to complete the installation of a remote starter.
- All connectors are displayed from the wire side (unless specified otherwise).
- Refer to the "Vehicle connections" following the installation diagram.

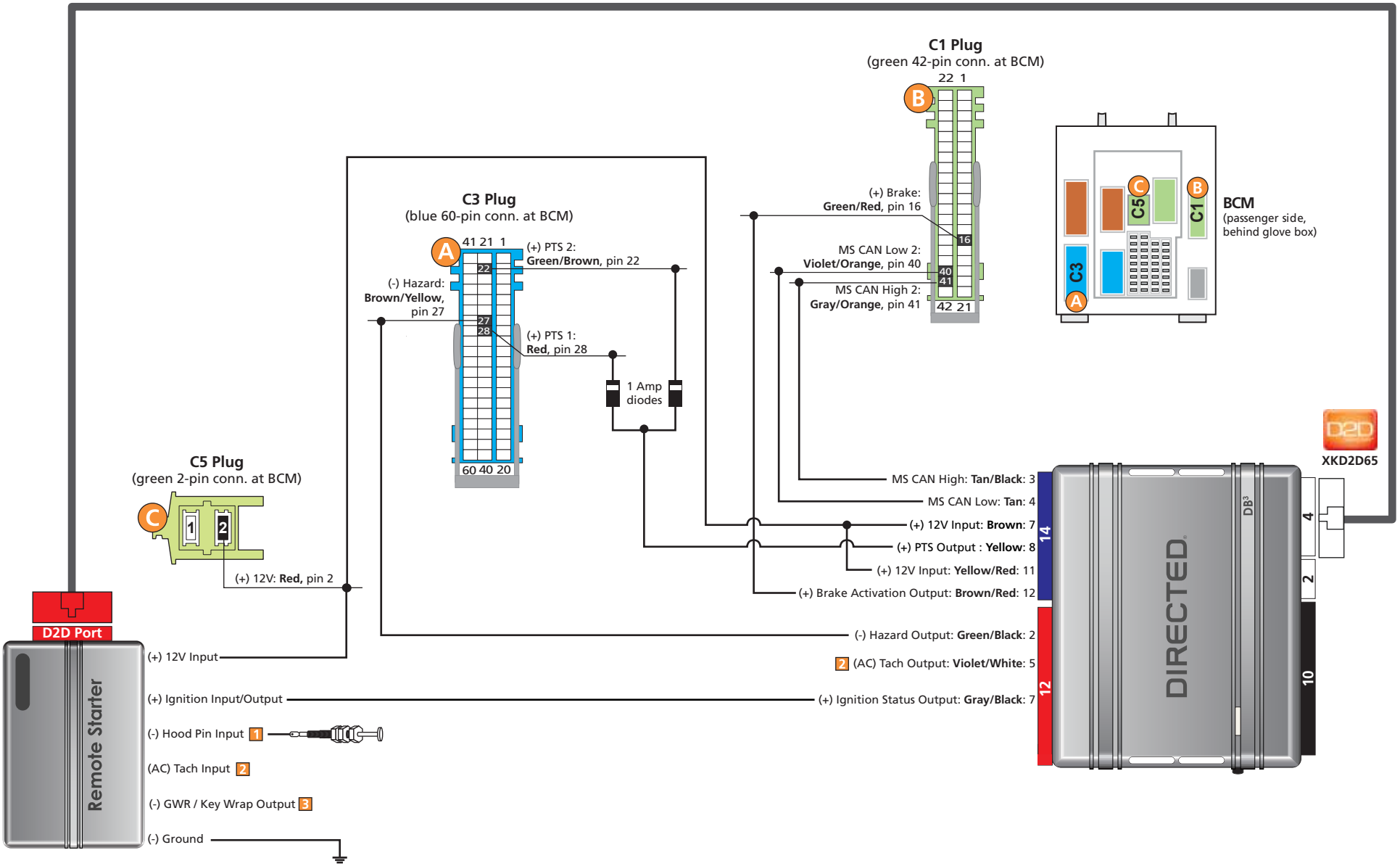
Footnotes 1 2 3

[1] **Hood pin**: The installation of an aftermarket hood pin is only required on vehicles that are not equipped with a factory hood pin or if not supported in data.
[2] **Tach:** An optional connection required on some remote starters, which do not support a tach signal in D2D.
[3] **Key wrap:** Refer to the Immobilizer interface section of the guide for additional information.

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Wiring diagram for vehicle equipped with an Automatic Transmission

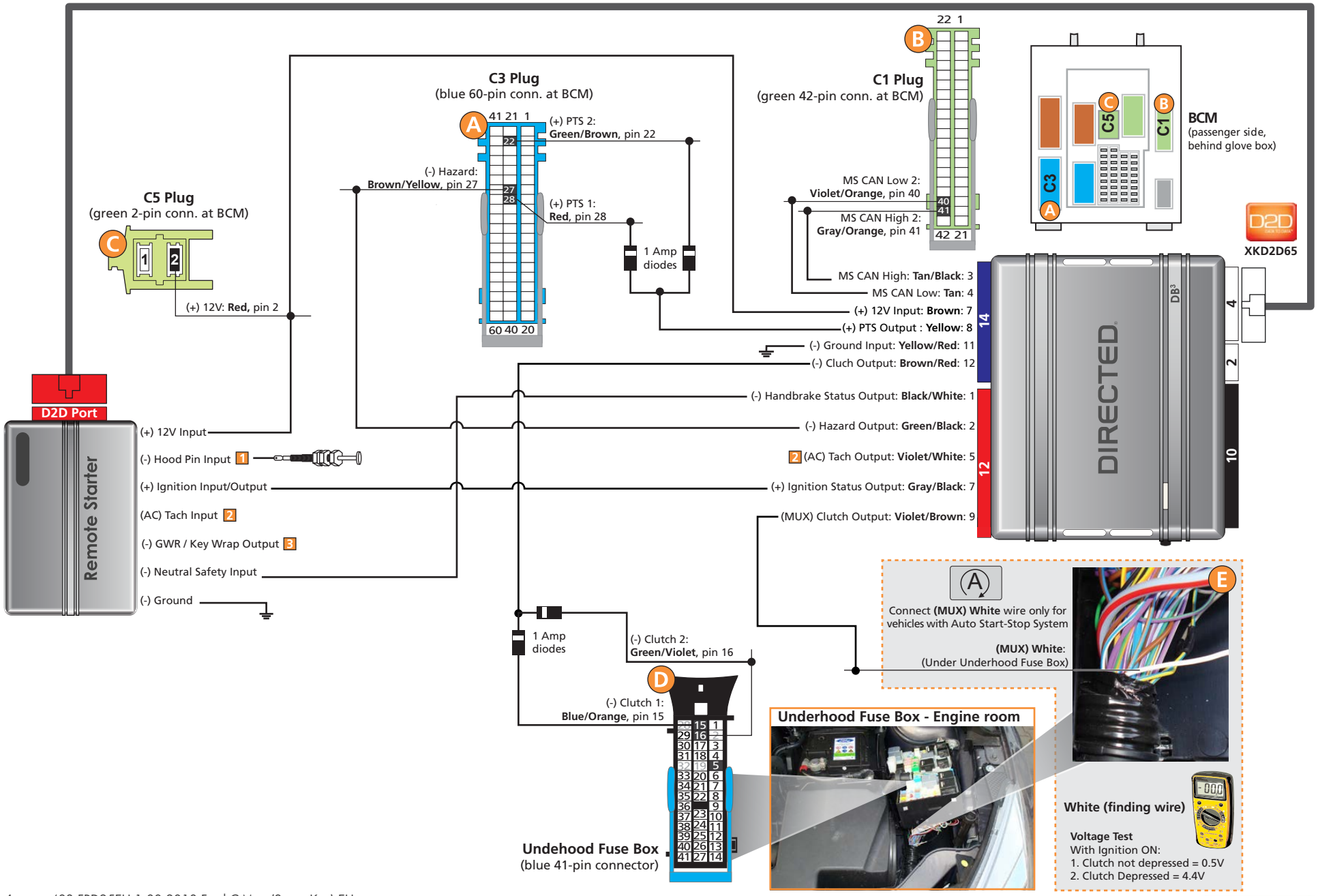
Refer to "Pre-installation and application warnings" for important information, such as the description of each special note referenced in the diagram (1 2 3).



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Wiring diagram for vehicle equipped with a Manual Transmission

Refer to "Pre-installation and application warnings" for important information, such as the description of each special note referenced in the diagram (1 2 3).



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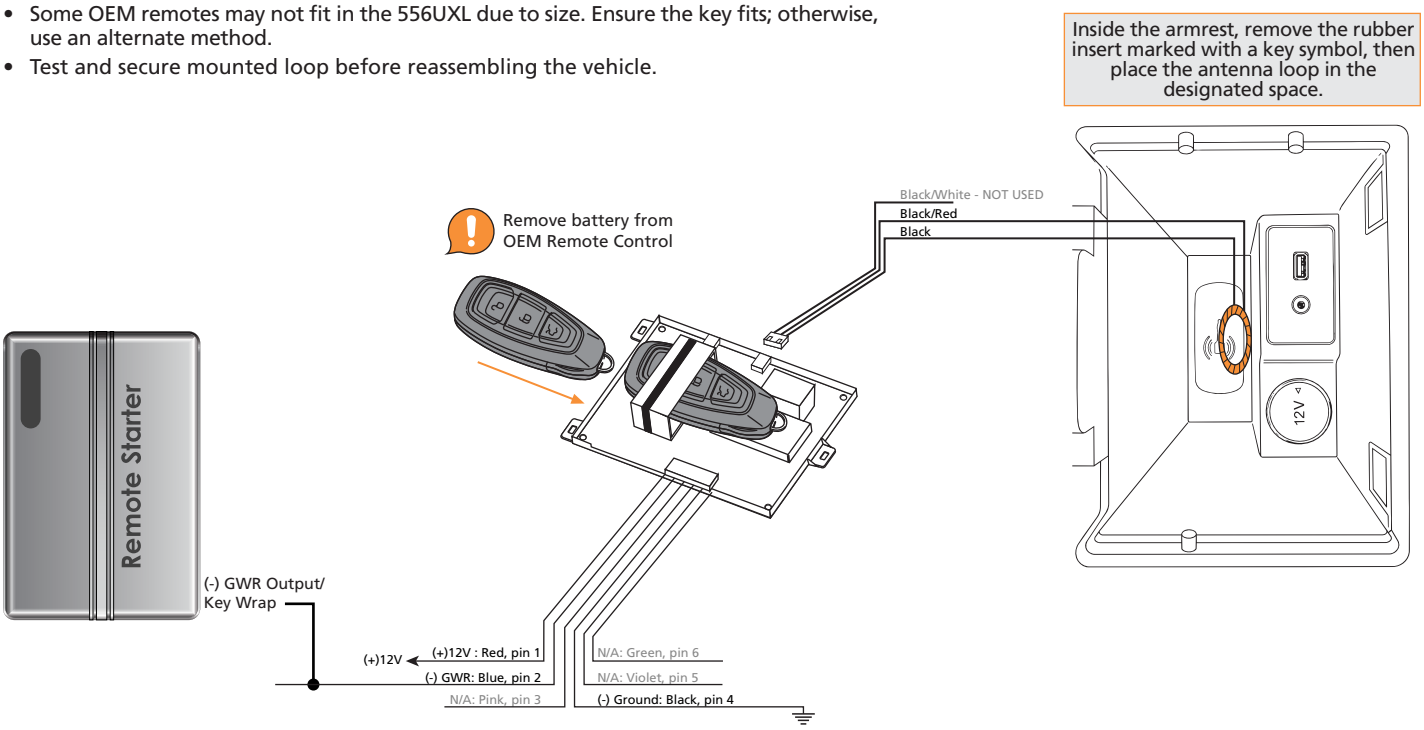
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Immobilizer interface 3

For vehicles with Keyless Entry, remove the battery from the OEM key (the OEM remote may differ from the models shown in the illustrations).

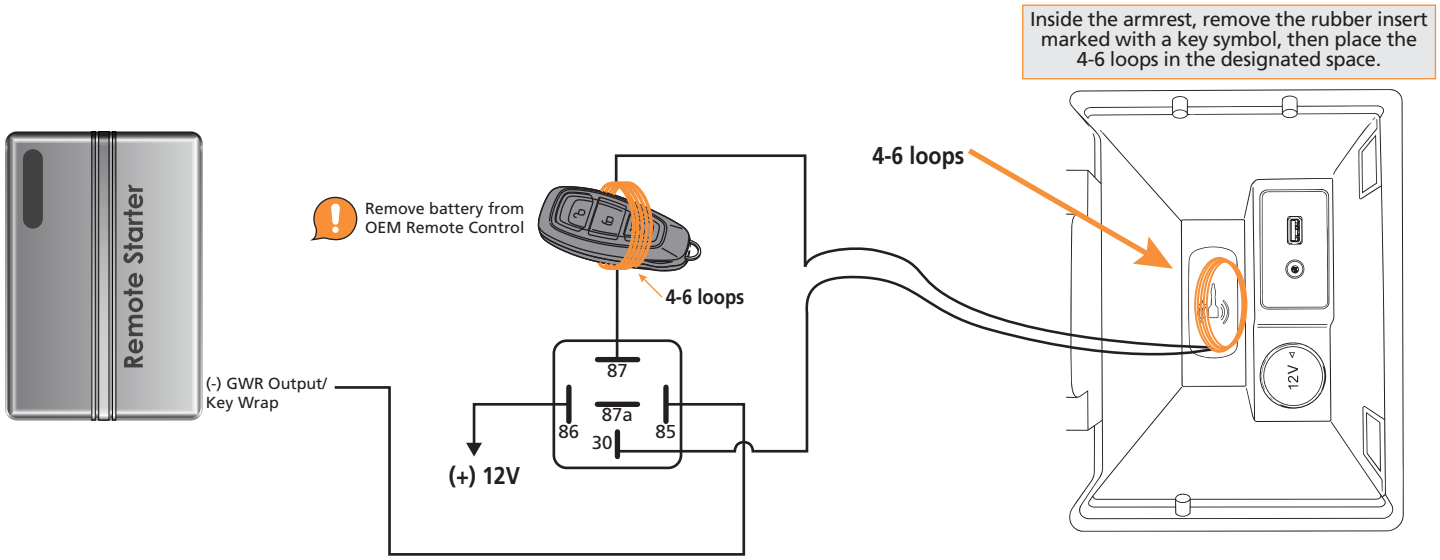
Method 1: Immobilizer interface using 556UXL (sold separately)

- Some OEM remotes may not fit in the 556UXL due to size. Ensure the key fits; otherwise, use an alternate method.
- Test and secure mounted loop before reassembling the vehicle.



Method 2: Loop immobilizer

- Make 4-6 loops and place the antenna loop as inside the armrest as shown below.
- Make 4-6 loops and place them around the front of the plastic body of the OEM key.
- Test and secure mounted loops before reassembling the vehicle.



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Locating components in the vehicle



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Vehicle connections

The connection information listed below is specific to the 2018 Ford C-Max (Smart Key) EU. Refer to "Pre-installation and application warnings" for a list of important notes.

Data 14-pin blue connector (H1)

Module			Connects To		
Conn./Pin	**Color**	**Description**	**Wire Location**	**(+/-)**	**Wire Color**
H1/1	Light Green	No Connection	No Connection		
H1/2	Violet/Yellow	No Connection	No Connection		
H1/3	Tan/Black	(DATA) HS CAN High	BCM behind glove box, C1 plug (green 42 pin), pin 41	data	gray/orange (MS CAN)
H1/4	Tan	(DATA) HS CAN Low	BCM behind glove box, C1 plug (green 42 pin), pin 40	data	violet/orange (MS CAN)
H1/5	Orange/Green	No Connection	No Connection		
H1/6	Orange/Brown	No Connection	No Connection		
H1/7	Brown	Relay N.O. (+) 12V input	BCM behind glove box, C5 plug (green 3 pin), pin 2	+	red (50A)
H1/8	Yellow	Relay COM. (+) PTS Output	BCM behind glove box, C3 plug (blue 60 pin), pin 28 BCM behind glove box, C3 plug (blue 60 pin), pin 22 Must use two 1A diodes.	+ +	red green/brown
H1/9	Orange/Yellow	Relay N.C. No Connection	No Connection		
H1/10	Orange/Red	Relay N.C. No Connection	No Connection		
H1/11	Yellow/Red	Relay COM. (+) 12V Input or (-) Ground Input	BCM behind glove box, C5 plug (green 3 pin), pin 2 OR	+	red (50A) (chassis ground)
H1/12	Brown/Red	Relay N.O. (-) Clutch Activation Output OR (+) Brake Activation Output	Underhood Fuse Box, blue 41 pin plug, pin 15 Underhood Fuse Box, blue 41 pin plug, pin 16 OR BCM behind glove box, C1 plug (green 42 pin), pin 16 Use Brake wire for Automatic Transmission or use Clutch for manual transmission.	- - +	blue/orange green/violet green/red
H1/13	*Red*	*W2W Only - (+) 12V input*	*Remote Starter*		
H1/14	*Black*	*W2W Only - (_) Ground*	*Remote Starter*		

Output 12-pin red connector (H2)

Module			Connects To		
Conn./Pin	**Color**	**Description**	**Wire Location**	**(+/-)**	**Wire Color**
H2/1	Black/White	(-) Handbrake Brake Status Output	Remote Starter		
H2/2	Green/Black	(-) Hazard Light Output	BCM behind glove box, C3 plug (blue 60 pin), pin 27	-	brown/yellow
H2/3	*Green/White*	*W2W Only - (-) Door Output*	*Remote Starter*		
H2/4	*Red/Black*	*W2W Only - (-) Trunk Output*	*Remote Starter*		
H2/5	*Violet/White*	*W2W only - (AC) Tach Output* [2]	*Remote Starter* *An optional connection required on some remote starters, which do not support a tach signal in D2D.*		
H2/6	Gray	No Connection	No Connection		
H2/7	Gray/Black	(+) Ignition Status Output	Remote Starter		
H2/8	Violet/Green	No Connection	No Connection		

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Module			Connects To		
Conn./Pin	Color	Description	Wire Location	(+/-)	Wire Color
H2/9	Violet/Brown	(MUX) Clutch Activation Output	under Underhood Fuse Box (only for vehicles with start stop system) Connection required for vehicles with manual transmission only.	MUX	white
H2/10	Yellow/Black	No Connection	No Connection		
H2/11	Orange/Black	No Connection	No Connection		
H2/12	Blue/Red	No Connection	No Connection		

Input 10-pin black connector (H3)

Module			Connects To		
Conn./Pin	Color	Description	Wire Location	(+/-)	Wire Color
H3/1	*Green*	*W2W Only - (-) Lock Input*	*Remote Starter*		
H3/2	*Blue*	*W2W Only - (-) Unlock Input*	*Remote Starter*		
H3/3	Red/White	No Connection	No Connection		
H3/4	White/Violet	No Connection	No Connection		
H3/5	Violet/Black	No Connection	No Connection		
H3/6	White/Black	No Connection	No Connection		
H3/7	*Pink/White*	*W2W Only - (+) Parking Light Input*	*Remote Starter*		
H3/8	*Violet*	*W2W Only - (+) Starter Input*			
H3/9	*Pink*	*W2W Only - (+) Ignition Input*	*Remote Starter*		
H3/10	*Blue/White*	*W2W Only - (-) Ground When Running (Status)*	*Remote Starter*		

D2D 4-pin white connector (H4)

Module		
Conn./Pin	Color	Description
H4/1	Blue	(Data) TX
H4/2	Black	(-) Ground
H4/3	Green	(Data) RX
H4/4	Red	(+) 12 Volt

RF 2-pin white connector (H5)

Module		
Conn./Pin	Color	Description
H5/1	N/A	No Connection
H5/2	N/A	No Connection

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Additional remote start connections

(Not all of these connections may be required for your installation. Refer to the wiring diagram for vehicle-specific connections.)

Wire	Connects To		
	Wire Location	(+/-)	Wire Color
12 Volts	BCM behind glove box, C5 plug (green 3 pin), pin 2	+	red (50A)
Ground			(chassis ground)
Hood Pin [1]	BCM behind glove box, C1 plug (green 42 pin), pin 15	- N.C.	blue/orange
IMMO Antenna - Must use a relay			(immobilizer antenna is in center console under tray)
Tachometer [3]	Not Applicable		

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Module programming

Refer to “LED diagnostics and troubleshooting” for more information and for troubleshooting purposes.

For Convenience-Only programming, see next section.

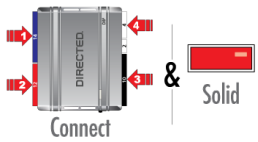
1 Ensure vehicle is in a safe location and cannot move forward during programming. For vehicles equipped with a manual transmission, make sure the gearshift is in the neutral position.



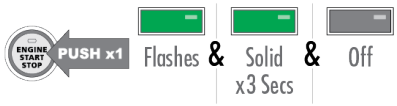
2 The harnesses must be connected in this exact order:

1. Connect the 14-pin.
2. Connect the 12-pin.
3. Connect the 10-pin.
4. Connect the 4-pin D2D cable (if being used).

When the LED turns ON solid red, proceed with the next step.



3 Turn the ignition ON.
The LED will flash green then stay ON green for 3 seconds, then turn OFF.



4 Turn the ignition OFF.



You have successfully completed the module programming sequence.

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Connecting the module

Important!

Make all the required connections to the vehicle, as described in this guide, and double-check to ensure everything is in order before moving onto the programming.

Warning! To take advantage of advanced features, you must use DirectLink 4.5 (and higher) or the Directechs Mobile application.

To flash a module using your computer:

1. Connect the interface module to your computer using the XKLoader2.
2. Go to *www.directechs.com* using Internet Explorer, and select **Flash Module**.
3. Follow the instructions to select your vehicle, installation type, and configure your options.
4. When you have configured the firmware options, click **FLASH**.

To flash a module using your smartphone or tablet:

1. Connect the interface module to your XKLoader3.
2. Launch the Directechs mobile app on your smartphone or tablet.
3. Select **FLASH YOUR MODULE** and follow the onscreen instructions.

When the flashing operation is successful, you can proceed with the instructions below.

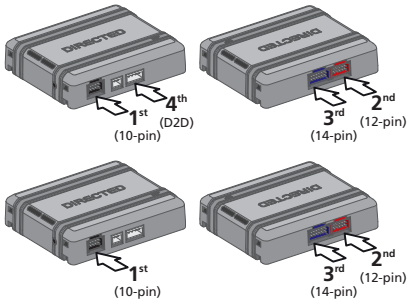
D2D installation

Recommended: Connect the 10-pin, 12-pin and 14-pin harnesses to the module, then connect the 4-pin D2D harness.

OR

W2W installation

If required: Connect the 10-pin and 12-pin harnesses to the module, then connect the 14-pin harness to the module.



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LED diagnostics and troubleshooting

This section provides LED diagnostics and troubleshooting information to guide you through the various stages of your installation.

Module Programming

LED	Description	Troubleshooting
Off	Module has no power.	Make sure the D2D harness is connected and that 12 Volt is present between the red and black wires. If 12 Volt is present, the module may be defective.
Solid red	Waiting to begin the programming sequence.	Ensure the correct programming procedure is being followed.
Flashes green & red	Initialization failed.	Reset the module and complete the programming again. If the issue persists, please contact Technical Support.
Solid orange	Transponder functions were skipped.	(If compatible) when RXT mode is not desired or convenience features are needed, please reset the and reprogram the module.
Flashes green	All required CAN networks has been detected.	Normal operation.
Flashes orange	1 of 2 CAN networks has been detected.	Normal operation.
Flashes orange slowly	Key2GO initiated.	Please follow the steps indicated in “Module programming” to complete the Key2GO programming.
Solid green x 3 secs	Module was successfully programmed with all functions.	Normal operation.
Solid orange x 3 secs	Module was successfully programmed without transponder functions.	Normal operation.

Module Programming - Error codes

LED	Description	Troubleshooting
Flashes red x 1	CAN2 not detected.	Check the CAN2 Orange/Green and Orange/Brown wire connections. Wake up the data bus by turning the ignition on and try again. If your installation does not require this connection, skip this step by pressing the programming button 5 times.
Flashes red x 1	J1850 not detected.	Check the J1850 wire connection. Wake up the data bus by turning the ignition on and try again.
Flashes red x 2	CAN1 not detected.	Check the CAN1 Tan and Tan/Black wire connections. Wake up the data bus by turning the ignition on and try again. If your installation does not require this connection, skip this step by pressing the programming button 5 times.
Flashes red x 3	Bypass data not detected.	Check the bypass line connection. If more than one wire is used, make sure they are not inverted. Ensure the vehicle still operates correctly using the factory key.
Flashes red x 4	Bypass processing error.	The bypass calculation failed. Reset the module and try again. If the condition persists, please contact Technical Support.
Flashes red x 5	ISO 1 not detected.	The Yellow/Black wire did not detect the expected signal. Refer to “Installation (wiring diagrams & vehicle wiring reference charts)” to check the connections.
Flashes red x 6	ISO 2 not detected.	The Orange/Black wire did not detect the expected signal. Refer to “Installation (wiring diagrams & vehicle wiring reference charts)” to check the connections.
Flashes red x 7	MUX not detected.	The Violet/Green or Violet/Brown wire did not detect the expected voltage value. Refer to “Installation (wiring diagrams & vehicle wiring reference charts)” to check the connections.

External Module Synchronization

LED	Description	Troubleshooting
(Flashes red, red, then orange) x 10	OBDII feature not supported.	The diagnostic data bus was not detected, therefore the SmartStart features will be limited.

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Active Ground When Running (Status)

LED	Description	Troubleshooting
Flashes green	Ground When Running (Status) command received.	The module has initialized the remote start sequence.
Flashes red & orange	Ignition ON command received.	The module has received the Ignition ON command and is processing the remote start sequence.
Flashes green quickly	Start ON command received.	The module has received the Start ON command and is processing the remote start sequence.
Flashes red x 10	PTS shutdown error.	The PTS output from the module was not activated due to safety protection.
Flashes red x 21	CAN bus incorrectly detected.	Verify the CAN1 and CAN2 connections. Refer to "Installation (wiring diagrams & vehicle wiring reference charts)" to check the connections.

Commands

LED	Description	Troubleshooting
Flashes orange x 1	LOCK command received.	If the bypass module fails to flash, it did not receive the signal. Commands can come from RF or D2D.
Flashes orange x 2	UNLOCK command received.	If the bypass module fails to flash, it did not receive the signal. Commands can come from RF or D2D.
Flashes orange x 3	TRUNK command received.	If the bypass module fails to flash, it did not receive the signal. Commands can come from RF or D2D.
Flashes orange x 4	AUX1 command received.	If the bypass module fails to flash, it did not receive the signal. Commands can come from RF or D2D.
Flashes orange x 5	AUX2 command received.	If the bypass module fails to flash, it did not receive the signal. Commands can come from RF or D2D.
Flashes orange x 6	AUX3 command received.	If the bypass module fails to flash, it did not receive the signal. Commands can come from RF or D2D.

Shutdown Codes

LED	Description	Troubleshooting
Flashes green x 1	Takeover successful.	Normal operation.
Flashes red x 1	Runsafe was not disabled.	No UNLOCK command was received prior to opening the door, or the 45 second timer expired in takeover mode.
Flashes red x 2	Brake was not detected.	The brakes were not detected, which prevents the system from shutting down the vehicle.
Flashes red x 3	Smart key was not detected.	The smart key was not detected, which prevents the system from shutting down the vehicle.
Flashes red x 4	Speed was detected.	The vehicle was detected as moving, which prevents the system from shutting it down.

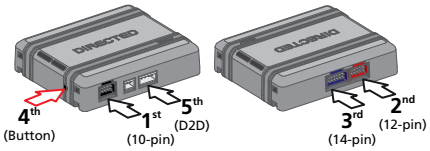
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Soft reset

A module reset will only erase the steps perfomed in “Module Programming”. The firmware and settings flashed to the module will not be affected.

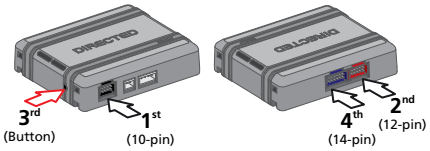
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D2D installation
Connect the 10-pin, 12-pin & 14-pin harnesses to the module.
Press and hold the programming button, then connect the 4-pin D2D harness.



OR

W2W installation
Connect the 10-pin & 12-pin harnesses to the module.
Press and hold the programming button, then connect the 14-pin harness to the module.



2 Wait 3 seconds until the LED turns ON solid orange then release the programming button. The LED turns ON solid red.



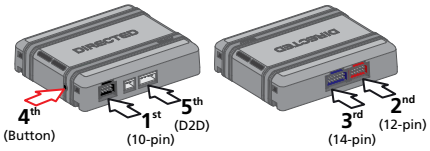
Hard reset

Warning Against Executing a Hard Reset!

A hard reset will revert the flashed firmware back to its default factory settings. Depending on the installation, some settings may need to be reconfigured. Connect your module to a computer and use the web configuration tool to edit its programmable features.

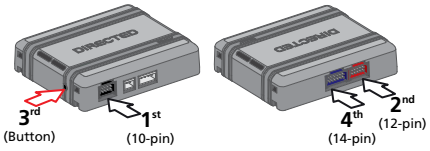
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D2D installation
Connect the 10-pin, 12-pin & 14-pin harnesses to the module. Press and hold the programming button, then connect the 4-pin D2D harness.



OR

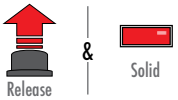
W2W installation
Connect the 10-pin & 12-pin harnesses to the module. Press and hold the programming button, then connect the 14-pin harness to the module.



2 Wait 3 seconds until the LED turns ON solid orange, and wait 10 more seconds until the LED starts to flash orange and red.



3 Release the programming button. The LED turns ON solid red.



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Feature programming

To enter feature programming routine:

1. Turn the ignition ON, then OFF.
2. Within 5 seconds, press and hold the programming button until the LED turns ON orange (after 3 seconds).
3. Release the programming button.
   The LED will flash green once slowly to indicate the feature number is 1. After a short delay, the LED flashes red rapidly to indicate the current option of feature 1 (e.g. 1 x green followed by 1 x red indicates feature 1 is set to option 1). The flashing sequence will repeat until a new command is entered.

Changing feature options:

4. Press the lock/arm or unlock/disarm button on the aftermarket transmitter to change the option of the selected feature.
   The LED flashes red rapidly the number of times equal to the current option number. After a short delay, the LED flashes green slowly the number of times to indicate the current feature. The flashing sequence will repeat until a new command is entered.

Accessing another feature:

4. Press and release the programming button a number of times to advance from the current feature to the next desired feature.
   The LED flashes green slowly the number of times equal to the feature number. After a short delay, the LED flashes red rapidly to indicate the current option of the current feature. The flashing sequence will repeat until a new command is entered. When the maximum number of features or options is reached, the LED will start flashing again from the first feature or option.

Exiting feature programming:

4. To exit feature programming, press and hold the programming button for 3 seconds or ensure there is no activity on the interface for 30 seconds. The LED will turn ON orange for 2 seconds to confirm the end of the programming sequence.

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Feature and option list

Depending on your vehicle, some of the features and options listed below may need to be configured using the configuration tool found on the module flashing page at www.directechs.com. The web offers more options; however, manual configuration of the features is possible using the information on this page.

*** Default option**

Feat.	Operation	Flashes / Option	Description
1	RFTD Output Type	**1. No RF Output***	Module is connected to a remote starter using a standard installation.
		2. RFTD Output	Module is connected to an XL202 using an RSR or RXT installation (when available).
		3. SmartStart	Module is connected to SmartStart using an RSR or RXT installation (when available).

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Limited 12 Months Consumer Warranty

DIRECTED BY VOXX LLC (the Company) warrants to the original purchaser of this product that should this product or any part thereof, under normal use and conditions, be proven defective in material or workmanship within 12 months from the date of original purchase, such defect(s) will be repaired or replaced with new or reconditioned product, (at the Company's option) without charge for parts or repair labor. A transmitter for a vehicle security or remote start system is warranted for 12 months from he date of original purchase.

To obtain repair or replacement within the terms of this Warranty, the product is to be delivered with proof of warranty coverage (e.g. dated bill of sale), authorization number, specification of defect(s), transportation prepaid, to an approved warranty station. This warranty is not transferable.

This Warranty does not cover damage to the vehicle's electrical system or costs incurred for the installation, removal or reinstallation of the product. This Warranty does not cover batteries, broken LCD transmitter display screens, nor apply to any product or part thereof which, in the opinion of the Company, has suffered or been damaged through alteration, improper installation, mishandling, misuse, abuse, neglect, accident, or by removal or defacement of the factory serial number/bar code label(s).

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Quick Reference Guide (customer copy)

Sending commands to your vehicle

Whether you want to remote start the engine, lock/unlock the doors or pop the trunk, there are 3 possible ways you can send commands to your vehicle, using the:

1. Factory remote.
2. Aftermarket remote.
3. Directed SmartStart® application via your smartphone.

If applicable, you can also start the engine remotely by pressing the Lock button 3 times quickly on your factory remote.

Vehicle takeover

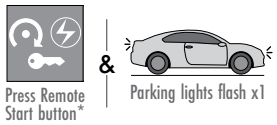
The vehicle takeover feature is **not** available in this vehicle. The engine will shut down as soon as a door is opened

Pit stop/idle mode

1 Stop the vehicle in a safe parking spot and put the gear in Park (P).

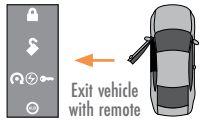
2 Press the Remote Start button on the transmitter.*

The parking lights will flash once to indicate the vehicle is now in Pit Stop Mode.



3 Exit the vehicle with the factory remote in hand (it is safe to leave the engine running).

Notes: We recommend that you always lock the doors of your vehicle when leaving it unattended.



* Icon and remote appearance may differ depending on the model purchased.

List of available commands

Note that the information below is for many Viper®, Clifford®, Python®, Avital®, Automate®, Autostart® and AstroStart® models. Icons and commands may differ depending on the model and options purchased. Refer to your authorized installation center for more specific information.

Button(s)	Actions
	Press & hold for 1 second to lock.
	Press & hold for 1 second to unlock.
*	Press & hold for 1 second to remote start.
AUX *	Press & hold for 5 seconds to activate the trunk release (optional).

* Icon and remote appearance may differ depending on the model purchased.

Notes

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